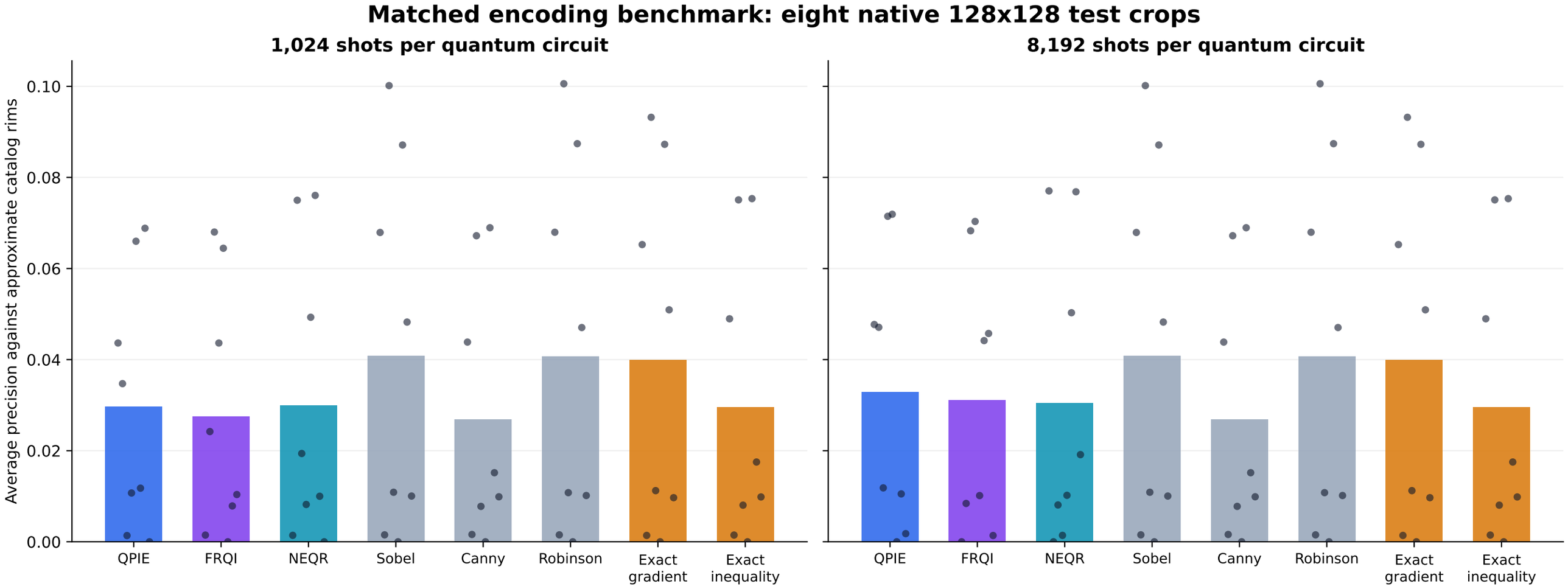
Matched encoding benchmark: eight native 128x128 test crops
1,024 shots per quantum circuit
8,192 shots per quantum circuit
Average precision against approximate catalog rims
0.10
0.08
0.06
0.04
0.02
0.00
QPIE
FRQI
NEQR
Sobel
Canny
Robinson
Exact gradient
Exact inequality
Bars: image macro means; dots: individual image means over three seeds. One empty-reference crop retained.